

Joe Buty

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EXPERIENCE

GenAI Data Engineer — Meta 2023 – Present | London, UK

- Identified opportunities and drove and/or owned the end-to-end development, performance evaluation, and data infrastructure for core large language model applications; partnered with engineering squads to deploy LLM/AIs (BERT, LLMs) to classify 60,000 daily active conversations by intent and urgency, improving the accuracy of automated AI resolutions and customer outcomes by 4%.
- Engineered clean, modular backend pipelines in Python and SQL for raw unstructured log collection, automated classifier scaling, and semantic data preparation to support production Retrieval-Augmented Generation (RAG), reducing data latency by 20% and improving context accuracy for AI responses.
- Maintained system-wide stability across complex distributed AI agent graphs by debugging state management capacity bottlenecks and building resilient fault tolerant and aware data ingestion loops.
- Translated granular engineering metrics, model validation, and pipeline constraints into clear, tactical technical roadmaps for non-technical cross-functional leaders and executive stakeholders.
- Created platform for human-expert to evaluate AI-Agent and LLM classification outcomes, increasing ground-truth annotation throughput by 30% and providing continuous feedback loops to proactively refine model performance.
- Led machine learning design reviews for predictive pipelines, guiding the architectural direction of models and ensuring engineering capabilities successfully addressed critical business targets
- Mentored junior engineers and cross-functional partners in LLM integration, unstructured data and privacy regulations, and robust development, accelerating individual and team technical proficiencies.

Data Engineer — Meta 2021 – 2023 | London, UK •

- Developed enterprise-scale, high-velocity ETL pipelines processing complex user workflows and interaction streams for millions of daily active advertisers using advanced SQL and Python scripting, enhancing data freshness by 40% to support real-time strategic decision-making.
- Transformed unstructured operational log data into automated performance scorecards, delivering transparent, audited data streams used to manage corporate vendor accountability and enforce strict contractual SLAs, saving an estimated \$6M annually in vendor inefficiencies.

Business Product Specialist — Facebook 2019 – 2021 | San Francisco, CA •

- Programmatically anonymized, structured, and securely transformed massive B2B communication text corporuses to build compliant, high-quality training and evaluation datasets for early language modeling initiatives.
- Owned ad creation support and directed a dotted-line workforce of 14 specialized support agents; interfaced with product stakeholders to identify high-severity bugs and confusions, rapidly resolving issues that protected over \$5M in at-risk quarterly ad revenue.
- Implemented an internal API translation service and built real-time data mining, ad buy spreadsheet integrations, and programmatic bulk data ingestion, increasing bulk ad buy creation success by 35%.
- Founded an operational data quality program to directly measure product support health and map performance data against top-line KPIs like customer churn and revenue.

Data Science Analyst — StubHub 2018 – 2019 | San Francisco, CA

- Engineered and deployed real-time low-latency predictive models using gradient boosted decision trees and logistic regressions directly inside high-volume transactional paths to intercept account fraud and lower chargeback liabilities and account takeover by over 20%.
- Programmatically integrated transactional risk-signal engine outputs with core web access software to

dynamically enforce multi-factor authentication (2FA) challenges for high-vulnerability client profiles.

Revenue Analyst & Contracting Work — Tech & Consulting Sector 2014 – 2018 | United States

- **Google:** partnered with commercial teams to translate deeply technical backend data architecture into high-level business narratives for corporate executives, bridging the gap between engineering metrics and executive decision-making.
- **Great Place to Work:** Designed and deployed interactive reporting dashboards and automated recommendation engines that empowered B2B clients to visualize workplace culture metrics, directly driving executive strategy and client retention.
- **KIPP Foundation:** Architected comprehensive data visualization and tracking systems to audit and optimize the allocation of a \$50M USD grant, ensuring rigorous compliance and clear impact reporting for institutional stakeholders.
- **The North Face:** Accelerated regional revenue performance analysis by synthesizing complex multi-channel sales logs into actionable CRM insights, allowing commercial leaders to rapidly adapt to shifting market trends.

TECHNICAL SKILLS MATRIX

Programming Languages: Python, SQL, JavaScript, PHP, Bash

AI Infrastructure: Data base architecture and governance, LLM integration accuracy and performance measurement, RAG system design and optimization, LLM Chaining (BERT), Agentic Graph Tooling

Data & Integrations: Apache Spark, REST APIs, ETL/ELT Pipelines, Hadoop, Relational Databases

EDUCATION

Columbia Engineering New York, NY *Applied Data Science Program*

California State University, Chico Chico, CA *Bachelor of Science (BS), Double Major: Mathematics (Statistics Focus) & Mechanical Engineering*